SUMMER HACK 2026 · HACKATHON PROJECT REPORT

MINDSCREEN

AI-Powered Burnout Detection System

Wearable signals · Facial emotion AI · Behavioral tracking · AXON v3 Assistant

TEAM

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THE PROBLEM

Wellness tools react late, because they wait for you to say something

Manual check-ins

Existing tools depend on the user remembering to self-report stress — easy to skip, easy to sugar-coat.

Late detection

Without continuous tracking, warning signs of burnout are only caught after they've already escalated.

Generic feedback

"Take a break" advice ignores what's actually happening with the user's body, mood, and workload.

MindScreen's answer: continuous, passive, and intelligent monitoring — glanceable throughout the day.

THE SOLUTION

Four passive signals, fused into one Burnout Score

Physical Activity

Steps & calories, synced from Google Fit (OAuth 2.0).

Emotional State

Facial expression read live from the webcam via face-api.js.

Behavioral Signals

Typing speed (WPM) and mouse-movement intensity, tracked passively.

Vital Stats

Heart rate (BPM) and SpO2, shown on an animated ECG-style pulse.

→ Burnout Score (0–100%) + AXON v3 tactical advice

SCORING SYSTEM

One number drives the entire HUD

0–100%

BURNOUT SCORE

Combines activity, mood, behavior, and vitals into a single live figure, recalculated as new signals arrive.

OPTIMAL

Score ≤ 50

Stable — system on standby.

WARNING

Score 51–80

Elevated stress detected.

CRITICAL

Score > 80

Immediate rest advised.

THE BRAIN

AXON v3 — a tactical AI assistant grounded in live data

Powered by Groq

Llama 3.1 8B Instant

Every reply is grounded in the user's live mood, burnout score, heart rate, and step count — injected straight into AXON's system prompt, capped at ~60 words for a snappy HUD feel.

Also reads its replies aloud via the Web Speech API.

QUICK COMMANDS

- > BIO_REPORT
- > STRESS_ANALYSIS
- > TACTICAL_ADVICE
- > SYSTEM_SCAN

FEATURE SET

What's inside the HUD

Facial Mood Detection

face-api.js reads happy / sad / angry / neutral / surprised, live.

Boost Mode

A high-confidence "surprised" spike auto-triggers a burst animation.

Passive Activity Tracking

Typing speed & mouse intensity, recomputed every 5 seconds.

Voice Feedback

AXON's replies are spoken aloud with the Web Speech API.

Cyberpunk HUD UI

Neon cyan/rose HUD, X-ray avatar panel, fully responsive.

Demo Profile Switcher

Five preset personas let judges preview every score band instantly.

UNDER THE HOOD

Tech Stack

React 19 + Vite 8	Frontend
Tailwind CSS 4	Styling
Groq · Llama 3.1 8B	AI Model
face-api.js	Face Detection
TensorFlow.js	On-device ML
Google Fit API	Wearables (OAuth 2.0)
Web Speech API	Voice
anime.js	Animation
Vercel	Deployment
Git & GitHub	Version Control

HOW IT WORKS

From signal to spoken advice

1 Connect

User links Google Fit (OAuth 2.0) or loads a demo profile.

2 Sense

Fit API returns steps/calories; webcam streams into face-api.js.

3 Track

Keystrokes & mouse activity build typing speed and movement.

4 Score

Signals combine into the Burnout Score and status band.

5 Reason

Live data is packed into AXON's prompt, sent to Groq/Llama.

6 Speak

AXON's answer renders in chat and is read aloud.

WHO IT'S FOR

Target Users & Impact

TARGET USERS

- Students
- Software developers
- Working professionals
- Fitness-tracker users
- Smartwatch companies

IMPACT

Surfaces early burnout signs before they escalate

Builds day-to-day productivity self-awareness

Nudges small, timely interventions over one-off check-ins

HONEST STATUS

Limitations & Future Scope

LIMITATIONS

- A wellness prototype, not a medical or diagnostic tool
- Facial-emotion accuracy depends on webcam & lighting
- Needs an existing Google Fit data source
- Heart rate / SpO_2 are simulated for the demo, not sensor-fed
- Burnout Score is a heuristic estimate, not clinically validated

FUTURE SCOPE

- Sleep tracking integration
- Multi-device support: Apple Health, Fitbit, Garmin
- Historical analytics dashboard
- Firebase authentication for real accounts
- React Native mobile companion app
- Guided meditation & breathing assistant

SYSTEM STANDING BY

THANK YOU

MindScreen — AI-Powered Burnout Detection System

Live Demo

mindscreen-one.vercel.app

Repository

github.com/yashtambade56-ux/Mindscreen

Full Report

Notion — MindScreen Hackathon Report

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